

B RANDOM MATRIX ANALYSIS OF THE GENERAL MODEL

In the generalized model, we consider that the synthetic data follow the following distribution:

$$\tilde{\mathbf{x}}_i \sim \mathcal{N}(\boldsymbol{\mu}_\beta, \mathbf{C}), \quad \boldsymbol{\mu}_\beta = \beta \boldsymbol{\mu} + \boldsymbol{\mu}^\perp \quad (13)$$

where $\beta \in \mathbb{R}$ defines the alignment of the synthetic mean with the mean of real data, and $\boldsymbol{\mu}^\perp$ is a vector orthogonal to $\boldsymbol{\mu}$.

Now we will analyze here the performance of the classifier given by equation (6), and prove a generalized theorem B.1 in the paper.

$$\mathbf{w}_q = \frac{1}{N} \mathbf{Q}(\gamma) \left(\mathbf{X} \mathbf{y} + \tilde{\mathbf{X}} \mathbf{D}(q) \tilde{\mathbf{y}} \right), \quad \mathbf{Q}(\gamma) = \left(\frac{1}{N} \mathbf{V} \mathbf{V}^\top + \gamma \mathbf{I}_p \right)^{-1}.$$

The performance of (6) are fully determined by the first two order moments: $\mathbb{E}[\mathbf{w}_q^\top \mathbf{x}]$ and $\mathbb{E}[(\mathbf{w}_q^\top \mathbf{x})^2]$.

B.1 TEST EXPECTATION:

We have that:

$$\mathbf{w}_q = \frac{1}{N} \sum_{i=1}^n \mathbf{Q} y_i \mathbf{x}_i + \frac{1}{N} \sum_{i=1}^m \mathbf{Q} q_i \tilde{y}_i \tilde{\mathbf{x}}_i$$

Let $\mathbf{x} \sim \mathcal{N}((-1)^a \boldsymbol{\mu}, \mathbf{I}_p)$ be a test sample independent of all the training samples $(\mathbf{v}_i)_{i=1}^N$. Then:

$$\mathbb{E}[\mathbf{w}_q^\top \mathbf{x}] = \frac{1}{N} \sum_{i=1}^n \mathbb{E}[y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x}] + \frac{1}{N} \sum_{i=1}^m \mathbb{E}[q_i \tilde{y}_i \tilde{\mathbf{x}}_i^\top \mathbf{Q} \mathbf{x}]$$

First sum: We have that, using the same lemma A.3:

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^n \mathbb{E}[y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x}] &= \frac{1}{N} \sum_{i=1}^n \frac{1}{1+\delta} \mathbb{E}[y_i \mathbf{x}_i^\top \mathbf{Q}_{-\mathbf{x}_i} \mathbf{x}] \\ &= \frac{1}{N} \sum_{i=1}^n \frac{1}{1+\delta} \mathbb{E}[\mathbf{x}_i]^\top \mathbb{E}[\mathbf{Q}_{-\mathbf{x}_i}] \mathbb{E}[\mathbf{x}] \\ &= \frac{1}{N} \sum_{i=1}^n \frac{(-1)^a}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} \\ &= \frac{(-1)^a \pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} \end{aligned}$$

Thus,

$$\frac{1}{N} \sum_{i=1}^n \mathbb{E}[y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x}] = \frac{(-1)^a \pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} \quad (14)$$

Second sum: Using the same lemma A.3:

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^m \mathbb{E}[q_i \tilde{y}_i \tilde{\mathbf{x}}_i^\top \mathbf{Q} \mathbf{x}] &= \frac{1}{N} \sum_{i=1}^m \frac{1}{1+\delta_S} \mathbb{E}[q_i \tilde{y}_i \tilde{\mathbf{x}}_i^\top \mathbf{Q}_{-\tilde{\mathbf{x}}_i} \mathbf{x}] \\ &= \frac{1}{N(1+\delta_S)} \sum_{i=1}^m \mathbb{E}[q_i \tilde{y}_i] \mathbb{E}[\tilde{\mathbf{x}}_i]^\top \mathbb{E}[\mathbf{Q}_{-\tilde{\mathbf{x}}_i}] \mathbb{E}[\mathbf{x}] \\ &= \frac{(-1)^a}{N(1+\delta_S)} \sum_{i=1}^m \lambda \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu} \\ &= \frac{(-1)^a \lambda (1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu} \end{aligned}$$

where (here y_i means the true label of $\tilde{\mathbf{x}}_i$):

$$\begin{aligned}\mathbb{E}[q_i \tilde{y}_i] &= y_i \mathbb{P}[q_i = 1 \mid \tilde{y}_i = y_i] - y_i \mathbb{P}[q_i = 1 \mid \tilde{y}_i \neq y_i] \\ &= y_i(\phi(1 - \varepsilon) - \rho\varepsilon) = \lambda y_i\end{aligned}$$

Therefore,

$$\mathbb{E}[\mathbf{w}_q^\top \mathbf{x}] = (-1)^a \left(\frac{\pi}{1 + \delta} \boldsymbol{\mu}^\top + \frac{\lambda(1 - \pi)}{1 + \delta_S} \boldsymbol{\mu}_\beta^\top \right) \bar{\mathbf{Q}} \boldsymbol{\mu} \quad (15)$$

B.2 TEST VARIANCE:

To determine the variance of $\mathbf{w}_q^\top \mathbf{x}$, it only remains to compute its second order. We have that:

$$\begin{aligned}\mathbb{E}[(\mathbf{w}_q^\top \mathbf{x})^2] &= \frac{1}{N^2} \mathbb{E} \left[\left(\sum_{i=1}^n y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} + \sum_{j=1}^m q_j \tilde{y}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q} \mathbf{x} \right)^2 \right] \\ &= \frac{1}{N^2} \mathbb{E} \left[\left(\sum_{i=1}^n y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \right)^2 \right] + \frac{1}{N^2} \mathbb{E} \left[\left(\sum_{j=1}^m q_j \tilde{y}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q} \mathbf{x} \right)^2 \right] + \frac{2}{N^2} \mathbb{E} \left[\left(\sum_{i=1}^n y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \right) \left(\sum_{j=1}^m q_j \tilde{y}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q} \mathbf{x} \right) \right]\end{aligned}$$

Let us compute each sum on its own, and then group the results at the end.

First sum: We have that:

$$\begin{aligned}\frac{1}{N^2} \mathbb{E} \left[\left(\sum_{i=1}^n y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \right)^2 \right] &= \frac{1}{N^2} \sum_{i=1}^n \sum_{k=1}^n \mathbb{E}[y_i y_k \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \mathbf{x}_k^\top \mathbf{Q} \mathbf{x}] \\ &= \frac{1}{N^2} \sum_{i \neq k} \mathbb{E}[y_i y_k \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \mathbf{x}_k^\top \mathbf{Q} \mathbf{x}] + \frac{1}{N^2} \sum_{i=1}^n \mathbb{E}[\mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \mathbf{x}_i^\top \mathbf{Q} \mathbf{x}]\end{aligned}$$

- For $i \neq k$, we have that:

$$\begin{aligned}\mathbb{E}[y_i y_k \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \mathbf{x}_k^\top \mathbf{Q} \mathbf{x}] &= \mathbb{E}[y_i y_k \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \mathbf{x}^\top \mathbf{Q} \mathbf{x}_k] \\ &= \mathbb{E}[y_i y_k \mathbf{x}_i^\top \mathbf{Q} \Sigma \mathbf{Q} \mathbf{x}_k] \\ &= \frac{1}{(1 + \delta)^2} \mathbb{E}[y_i y_k \mathbf{x}_i^\top \mathbf{Q}_{-\mathbf{x}_i} \Sigma \mathbf{Q}_{-\mathbf{x}_k} \mathbf{x}_k] \\ &= \frac{1}{(1 + \delta)^2} \mathbb{E} \left[y_i y_k \mathbf{x}_i^\top \left(\mathbf{Q}_{-\mathbf{x}_{i,k}} - \frac{\frac{1}{N} \mathbf{Q}_{-\mathbf{x}_{i,k}} \mathbf{x}_k \mathbf{x}_k^\top \mathbf{Q}_{-\mathbf{x}_{i,k}}}{1 + \delta} \right) \Sigma \left(\mathbf{Q}_{-\mathbf{x}_{i,k}} - \frac{\frac{1}{N} \mathbf{Q}_{-\mathbf{x}_{i,k}} \mathbf{x}_i \mathbf{x}_i^\top \mathbf{Q}_{-\mathbf{x}_{i,k}}}{1 + \delta} \right) \mathbf{x}_k \right] \\ &= \frac{1}{(1 + \delta)^2} (A_1 - A_2 - A_3 + A_4)\end{aligned}$$

And we have that:

$$\begin{aligned}A_1 &= \mathbb{E}[y_i y_k \mathbf{x}_i^\top \mathbf{Q}_{-\mathbf{x}_{i,k}} \Sigma \mathbf{Q}_{-\mathbf{x}_{i,k}} \mathbf{x}_k] \\ &= \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu} \\ &= \boldsymbol{\mu}^\top \left(\frac{1 - b_2}{h} \bar{\mathbf{Q}} \Sigma \bar{\mathbf{Q}} + \frac{b_1}{h} \bar{\mathbf{Q}} \Sigma_\beta \bar{\mathbf{Q}} \right) \boldsymbol{\mu}\end{aligned}$$

And:

$$\begin{aligned}A_2 &= \frac{1}{N(1 + \delta)} \mathbb{E}[y_i y_k \mathbf{x}_i^\top \mathbf{Q}_{-\mathbf{x}_{i,k}} \mathbf{x}_k \mathbf{x}_k^\top \mathbf{Q}_{-\mathbf{x}_{i,k}} \Sigma \mathbf{Q}_{-\mathbf{x}_{i,k}} \mathbf{x}_k] \\ &= \frac{\text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])}{N(1 + \delta)} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}\end{aligned}$$

Since, by concentration laws:

$$\begin{aligned}
\frac{1}{N} \mathbf{x}_k^\top \mathbf{Q}_{-\mathbf{x}_{i,k}} \Sigma \mathbf{Q}_{-\mathbf{x}_{i,k}} \mathbf{x}_k &= \frac{1}{N} \mathbb{E}[\mathbf{x}_k^\top \mathbf{Q}_{-\mathbf{x}_{i,k}} \Sigma \mathbf{Q}_{-\mathbf{x}_{i,k}} \mathbf{x}_k] \\
&= \frac{1}{N} \mathbb{E}[\text{Tr}(\mathbf{x}_k \mathbf{x}_k^\top \mathbf{Q}_{-\mathbf{x}_{i,k}} \Sigma \mathbf{Q}_{-\mathbf{x}_{i,k}})] \\
&= \frac{1}{N} \text{Tr}(\mathbb{E}[\mathbf{x}_k \mathbf{x}_k^\top \mathbf{Q}_{-\mathbf{x}_{i,k}} \Sigma \mathbf{Q}_{-\mathbf{x}_{i,k}}]) \\
&= \frac{1}{N} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q}_{-\mathbf{x}_{i,k}} \Sigma \mathbf{Q}_{-\mathbf{x}_{i,k}}]) \\
&= \frac{1}{N} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])
\end{aligned}$$

And we can easily verify that:

$$A_3 = A_2, \quad A_4 = \mathcal{O}(N^{-1})$$

Thus,

$$\begin{aligned}
\frac{1}{N^2} \sum_{i \neq k} \mathbb{E}[y_i y_k \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \mathbf{x}_k^\top \mathbf{Q} \mathbf{x}] &= \frac{n^2 - n}{N^2} \left(\boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu} - \frac{2 \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])}{N(1 + \delta)} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} \right) \\
&= \frac{\pi^2}{(1 + \delta)^2} \left(\boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu} - \frac{2 \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])}{N(1 + \delta)} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} \right)
\end{aligned}$$

- And then, for $i \in \{1, \dots, n\}$:

$$\begin{aligned}
\mathbb{E}[\mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \mathbf{x}_i^\top \mathbf{Q} \mathbf{x}] &= \mathbb{E}[\mathbf{x}_i^\top \mathbf{Q} \Sigma \mathbf{Q} \mathbf{x}_i] \\
&= \frac{1}{(1 + \delta)^2} \mathbb{E}[\mathbf{x}_i^\top \mathbf{Q}_{-\mathbf{x}_i} \Sigma \mathbf{Q}_{-\mathbf{x}_i} \mathbf{x}_i] \\
&= \frac{1}{(1 + \delta)^2} \text{Tr}(\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top \mathbf{Q}_{-\mathbf{x}_i} \Sigma \mathbf{Q}_{-\mathbf{x}_i}]) \\
&= \frac{1}{(1 + \delta)^2} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])
\end{aligned}$$

Thus:

$$\frac{1}{N^2} \sum_{i=1}^n \mathbb{E}[\mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \mathbf{x}_i^\top \mathbf{Q} \mathbf{x}] = \frac{\pi}{N(1 + \delta)^2} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])$$

Hence, the first sum gives us:

$$\frac{1}{N^2} \mathbb{E} \left[\left(\sum_{i=1}^n y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \right)^2 \right] = \frac{\pi^2}{(1 + \delta)^2} \left(\boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu} - \frac{2 \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])}{N(1 + \delta)} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} \right) + \frac{\pi}{N(1 + \delta)^2} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) \quad (16)$$

Second sum: We have that:

$$\begin{aligned}
\frac{1}{N^2} \mathbb{E} \left[\left(\sum_{i=1}^m q_i \tilde{y}_i \tilde{\mathbf{x}}_i^\top \mathbf{Q} \mathbf{x} \right)^2 \right] &= \frac{1}{N^2} \sum_{i,j=1}^m \mathbb{E}[q_i q_j \tilde{y}_i \tilde{y}_j \tilde{\mathbf{x}}_i^\top \mathbf{Q} \mathbf{x} \mathbf{x}_j^\top \mathbf{Q} \mathbf{x}] \\
&= \frac{1}{N^2} \sum_{i \neq j} \mathbb{E}[q_i q_j \tilde{y}_i \tilde{y}_j \tilde{\mathbf{x}}_i^\top \mathbf{Q} \mathbf{x} \tilde{\mathbf{x}}_j^\top \mathbf{Q} \mathbf{x}] + \frac{1}{N^2} \sum_{i=1}^m \mathbb{E}[q_i \tilde{\mathbf{x}}_i^\top \mathbf{x} \tilde{\mathbf{x}}_i^\top \mathbf{Q} \mathbf{x}]
\end{aligned}$$

- For $i \neq j \in \{1, \dots, m\}$, we have that:

$$\begin{aligned}
\mathbb{E}[q_i q_j \tilde{y}_i \tilde{y}_j \tilde{\mathbf{x}}_i^\top \mathbf{Q} \mathbf{x} \tilde{\mathbf{x}}_j^\top \mathbf{Q} \mathbf{x}] &= \mathbb{E}[q_i q_j \tilde{y}_i \tilde{y}_j \tilde{\mathbf{x}}_i^\top \mathbf{Q} \Sigma \mathbf{Q} \mathbf{x}_j] \\
&= \frac{1}{(1 + \delta_S)^2} \mathbb{E}[q_i q_j \tilde{y}_i \tilde{y}_j \tilde{\mathbf{x}}_i^\top \mathbf{Q}_{-\tilde{\mathbf{x}}_i} \Sigma \mathbf{Q}_{-\tilde{\mathbf{x}}_j} \mathbf{x}_j] \\
&= \frac{1}{(1 + \delta_S)^2} \mathbb{E} \left[q_i q_j \tilde{y}_i \tilde{y}_j \tilde{\mathbf{x}}_i^\top \left(\mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}} - \frac{\frac{1}{N} \mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}} q_j \tilde{\mathbf{x}}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}}}{1 + \delta_S} \right) \Sigma \left(\mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}} - \frac{\frac{1}{N} \mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}} q_i \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top \mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}}}{1 + \delta_S} \right) \tilde{\mathbf{x}}_j \right] \\
&= \frac{1}{(1 + \delta_S)^2} (A_1 - A_2 - A_3 + A_4)
\end{aligned}$$